E-113
BPA / MP STEAM GENERATOR (KETTLE)
4.9 MW, 223 m², TEMA AKT

P-106A/B
TPA PUMPAROUND
293 m³/h @ 130 m
110 kW, API 610 OH2

P-107A/B
MPA PUMPAROUND
387 m³/h @ 138 m
160 kW, API 610 BB2

P-108A/B
BPA PUMPAROUND
318 m³/h @ 142 m
132 kW, API 610 BB2

C-101
ATMOSPHERIC CRUDE FRACTIONATOR
ID 6.9 m / 4.1 m strip. x 40.5 m T/T
41 valve trays (2/4-pass)

C-102
KERO SIDE STRIPPER
ID 1.7 m x 9.2 m T/T
6 valve trays

P-112A/B
ATM. RESIDUE / VACUUM HEATER CHARGE
398 m³/h @ 172 m
250 kW, API 610 BB2

C-103
DIESEL SIDE STRIPPER
ID 1.9 m x 9.2 m T/T
6 valve trays

C-104
AGO SIDE STRIPPER
ID 1.4 m x 9.2 m T/T
6 valve trays

P-109A/B
KERO PRODUCT
122 m³/h @ 148 m
55 kW, API 610 BB2

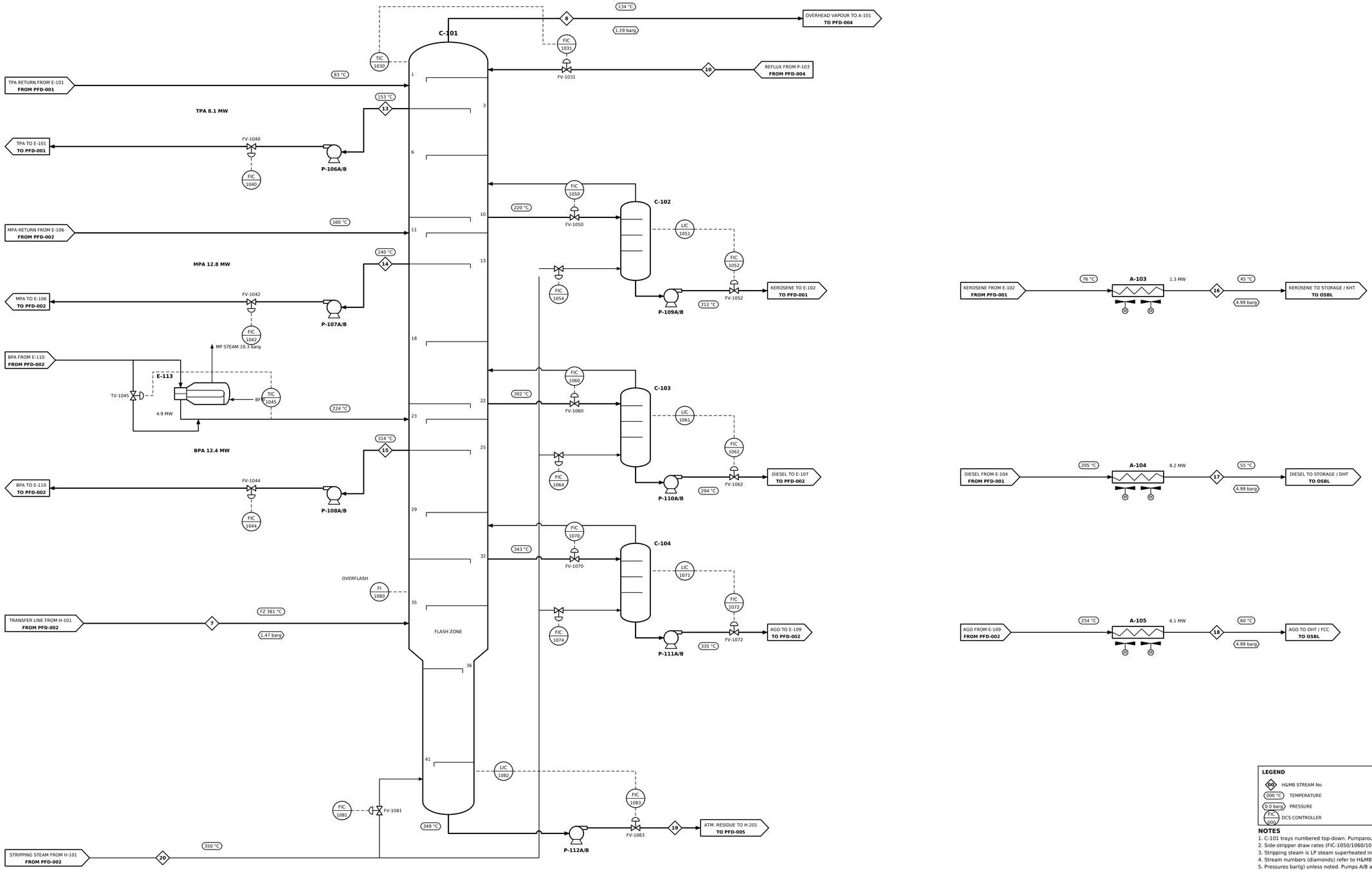
P-110A/B
DIESEL PRODUCT
151 m³/h @ 154 m
75 kW, API 610 BB2

P-111A/B
AGO PRODUCT
84 m³/h @ 153 m
45 kW, API 610 BB2

A-103
KERO COOLER
1.3 MW, 184 m² bare
1 bay(s), 2 fans x 11 kW

A-104
DIESEL COOLER
8.2 MW, 430 m² bare
1 bay(s), 2 fans x 45 kW

A-105
AGO COOLER
6.1 MW, 287 m² bare
1 bay(s), 2 fans x 30 kW



STREAM No.	7	8	10	13	14	15	16	17	18	19	20
DESCRIPTION	Heater outlet / transfer line	Column overhead vapour	Reflux	TPA pumparound (draw)	MPA pumparound (draw)	BPA pumparound (draw)	Kerosene product	Diesel product	AGO product	Atmospheric residue	Atm stripping steam (total)
PHASE	MIXED	VAPOUR	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	LIQUID	VAPOUR
TOTAL FLOW kg/h	569,197	229,224	101,310	166,764	208,349	165,727	68,595	81,613	45,847	256,909	11,681
TEMPERATURE °C	366	134	45	153	240	314	45	55	60	349	350
PRESSURE barg	2.4	1.2	7.0	8.0	8.0	8.0	5.0	5.0	5.0	19.0	3.5
STD LIQ. FLOW BPSD	100,000	45,042	21,105	33,584	39,539	29,853	13,017	14,701	7,930	40,413	-
VAPOUR FRACTION (mass)	0.533	1	0	0	0	0	0	0	0	0	1
MOLECULAR WEIGHT	215.7	97.9	97.9	114.2	158.6	216.5	158.6	216.5	280.8	528.6	-

LEGEND

- ◇00 H&MB STREAM No.
- (000 °C) TEMPERATURE
- (0.0 barg) PRESSURE
- FIC 000 DCS CONTROLLER
- INSTRUMENT SIGNAL
- CONTROL VALVE
- SERVICE **TO PFD-00X** OFF-SHEET CONNECTOR
- MAIN PROCESS
- SECONDARY PROCESS
- UTILITY
- FLUE GAS / AIR

NOTES

1. C-101 trays numbered top-down. Pumparound heat recovered in preheat trains (PFD-001/002).
2. Side-stripper draw rates (FIC-1050/1060/1070) set cut points; product rate on stripper level.
3. Stripping steam is LP steam superheated in H-101 convection (stream 20).
4. Stream numbers (diamonds) refer to H&MB CFU-000-PR-HMB-001; data from design case, normal operation.
5. Pressures bar(g) unless noted. Pumps A/B are 2 x 100 %; only duty pump shown.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PROCESS FLOW DIAGRAM
ATMOSPHERIC COLUMN C-101, PUMPAROUNDS & SIDE STRIPPERS

DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	3 OF 6

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER **CFU-100-PR-PFD-003** REV **A**